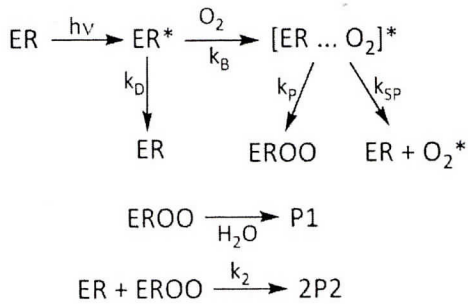
Physical Chemistry Cume Dec 1 2017

This cume is based upon the following announced paper:

Effects of Temperature and Concentration on the Rate of Photobleaching of Erythrosine in Water. Joshua K. G. Karlsson, Owen J. Woodford, Roza Al-Aqar, and Anthony Harriman. Journal of Physical Chemistry A (2017) Volume 121, Issue 45, 8569–8576.

DOI: 10.1021/acs.jpca.7b06440

1. Scheme 1 was given as:



And presented together with the following equations:

$$I_{ABS} = 1 - 10^{-A} \quad (2)$$

$$P_O = \frac{k_B[O_2]}{k_B[O_2] + k_D} \quad (3)$$

$$P_\Delta = \frac{\Phi_\Delta}{P_T} \quad (4)$$

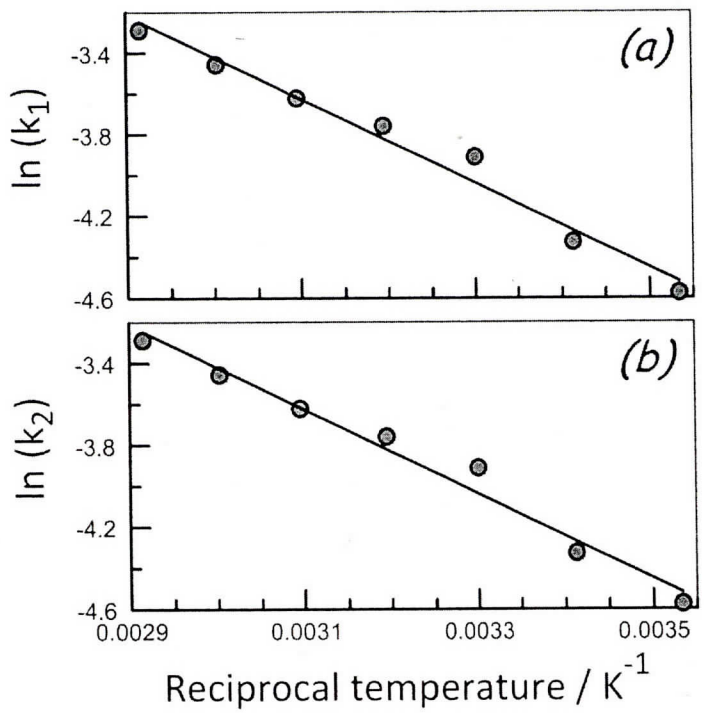
$$k_F = (I_A I_{ABS} \gamma) P_T P_O \kappa \quad (5)$$

$$P_Q = \frac{k_Q[ER]}{k_Q[ER] + k_\Delta} \quad (6)$$

Describe what each of these equations represent (i.e. describe what the parameters I_{ABS}, P_O, P_D, k_F and P_Q mean in terms of the mechanism).

2. The paper discusses at length the activation parameters for the inherent photobleaching process and the self-catalyzed photobleaching. This is based on table 2 and Figure 4 (see next page):

$[ER]_0$ (μM)	T (°C)	$[O_2]$ (μM)	k_1 (min^{-1})	k_2 (nM^{-1} min^{-1})
8.1	20	0.232	0.0093	0.39
19.5	10	0.285	0.0049	0.31
19.5	20	0.232	0.0069	0.46
19.5	30	0.196	0.0082	0.63
19.5	40	0.172	0.0091	0.82
19.5	50	0.156	0.0122	0.92
19.5	60	0.145	0.0137	1.01
19.5	70	0.138	0.0172	1.34



The calculated activation energy for the inherent process was 17.1 kJ/mol and for the self-catalyzed process was 18.5 kJ/mol. The authors also note that a previous study showed that diffusional addition of singlet molecular oxygen across double bonds in simple alkenes involves an activation energy close to zero. How do the authors relate these results to scheme 1 and the parameters in equations (2)-(6)?

3. Inherent photobleaching is a first order process while autocatalysis is a bimolecular process. Autocatalysis is the process whereby a product formed from the inherent process accelerates the reaction. Assuming that reactant converts to the autocatalytic product with a 1:1 stoichiometery, show that the relationship between reaction rate, r and reactant concentration c is given by:

$$r/c = (k_1 + k_2c_0) - k_2c$$

where k_1 is the inherent photobleaching rate constant, k_2 is the autocatalyzed rate constant and c_0 is the initial reactant concentration.